

Spectroscopy of ^{171}Yb in an optical lattice based on laser linewidth transfer using a narrow linewidth frequency comb

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Abstract: We propose a novel, high-performance, and practical laser source system for optical clocks. The laser linewidth of a fiber-based frequency comb is reduced by phase locking a comb mode to an ultrastable master laser at 1064 nm with a broad servo bandwidth. A slave laser at 578 nm is successively phase locked to a comb mode at 578 nm with a broad servo bandwidth without any pre-stabilization. Laser frequency characteristics such as spectral linewidth and frequency stability are transferred to the 578-nm slave laser from the 1064-nm master laser. Using the slave laser, we have succeeded in observing the clock transition of ^{171}Yb atoms confined in an optical lattice with a 20-Hz spectral linewidth.

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1. Introduction

Lasers with high frequency stability and a narrow linewidth are indispensable tools in various scientific fields. With optical clocks, lasers locked to a longitudinal mode of an ultrastable cavity are used to drive narrow-linewidth clock transitions in atoms/ions with a sub-hertz linewidth and a typical frequency stability of 10^{-15} with a 1-s averaging time [1]. Recently, a state of the art laser stabilized to a cavity was reported that had a stability of 10^{-16} with 1-s averaging [2]. To observe the clock transition in atoms, a high-finesse cavity with a coating that matches the wavelength of the atomic transition is usually used to narrow the laser linewidth. Therefore, multiple ultrastable cavities may be needed in experiments involving different clock transitions. On the other hand, at present, the short term frequency stability of optical clocks is limited by the Brownian motion in ultrastable cavities [3]. To overcome the limitation, two Sr optical lattice clocks were compared using lasers stabilized to a common cavity with extremely high frequency stabilities [4]. The frequency fluctuation of the cavity was largely canceled out in this scheme. When comparing optical clocks with quite different clock wavelengths using this scheme, we have to link the frequencies of the clocks by using an optical frequency comb. Although experimental studies of frequency stability transfer between two lasers operating at different wavelengths using an optical frequency comb have already been reported [5, 6], these frequency combs did not transfer the spectral linewidth on its own due to the limited servo bandwidth of the combs. Consequently, in these studies, extra ultrastable cavities were used for the clock lasers at the clock wavelengths to ensure that the slave laser has a narrow linewidth.

The simplest and most powerful way to increase the servo bandwidth of the entire transfer system is to increase the servo bandwidths of the frequency comb. High-speed controllable fiber-based frequency combs with an intracavity electro-optic modulator (EOM) have been reported [7–11]. These combs can efficiently transfer not only frequency stability but also the laser spectral linewidth to other wavelengths by phase locking the laser to a comb mode without any extra ultrastable cavity. In addition, we have already demonstrated magneto-optical trapping requiring a narrow spectral linewidth (few kHz level) by using a similar linewidth transfer system with an external cavity laser diode (ECLD) as the slave laser without any extra cavity [12]. In this paper, we propose a new clock laser system for spectroscopy at the several tens of Hz level by combining a narrow linewidth frequency comb and a 1064-nm ultrastable master laser. The 1064-nm Nd:YAG laser used in the system is appropriate as an ultrastable master laser because of its low intrinsic laser noise and outstanding performance [13, 14]. Using this system, we also demonstrate the spectroscopy of the $^1S_0 - ^3P_0$ clock transition in ^{171}Yb confined in an optical lattice at 578 nm. To the best of our knowledge, this is the first report of atomic spectroscopy based on a laser linewidth transfer scheme using just a narrow linewidth frequency comb.

2. Experimental setup and ultra-high resolution spectroscopy

The experimental setup is shown in Fig. 1. As a master laser yielding a narrow spectrum linewidth and high frequency stability, we used a Nd:YAG laser emitting at 1064 nm that was locked by the Pound-Drever-Hall technique [15] to the TEM₀₀ mode of an ultrastable cavity at 1064 nm by feedback to a piezoelectric transducer (PZT) bonded to the laser crystal in the Nd:YAG. The servo bandwidth was approximately 100 kHz. The 1064-nm cavity was a

Fabry-Pérot etalon made of ultra-low expansion glass with a thermal expansion coefficient of zero at around 21.6 °C. A local oscillator with a frequency of approximately 5 MHz drove the EOM to provide the light with sidebands. We were able to obtain a finesse of $\sim 735,000$ for the TEM₀₀ mode of the cavity. The length and diameter of the spacer were 75 and 25.4 mm, respectively. The Fabry-Pérot etalon was mounted vertically [13] in an aluminum vacuum chamber at a pressure of 3×10^{-5} Pa.

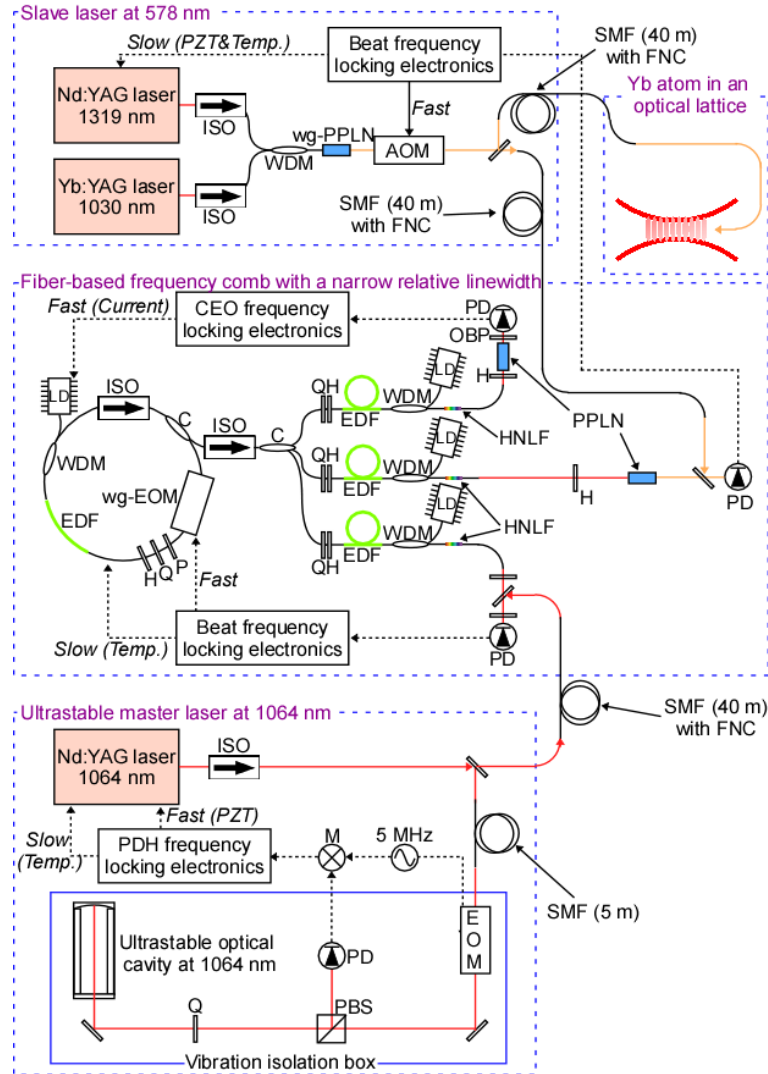


Fig. 1. Experimental setup of a clock laser system for an Yb optical lattice clock using linewidth transfer with a narrow linewidth comb; ISO, optical isolator; LD, laser diode for pumping; WDM, wavelength division multiplexing coupler; C, fiber coupler; P, polarizer; Q, quarter wave plate; H, half wave plate; EDF, erbium-doped fiber; OBP, optical bandpass filter; PD, photo detector; HNL, highly nonlinear fiber; M, frequency mixer; FNC, fiber noise canceling. AOMs for FNCs are omitted from the figure.

Room temperature vulcanizing silicone was used to bond the optical cavity to an aluminum disc, which was supported by three Teflon posts. The spectral linewidth of the ultrastable laser was estimated to be ~ 2 Hz by observing the beat note between the laser and another ultrastable laser. The highly stable 1064-nm laser beam was delivered to a frequency comb via a 40-m noise-cancelled single-mode fiber (SMF). The comb was based on an

erbium-doped fiber laser (repetition rate (f_{rep}): 43 MHz) with an intracavity waveguide EOM (wg-EOM) [10]. Three-branched erbium-doped fiber amplifiers were used to detect the beat note between the 1064-nm reference laser and the comb, the beat note between a 578-nm slave laser (mentioned below) and the comb, and the carrier envelope offset frequency (f_{CEO}). The first beat note at 1064 nm, and the f_{CEO} were phase-locked to an RF reference (30 MHz) by feedback to the wg-EOM, and the injection current of the pump laser for the mode-locked fiber laser, respectively. When all the phase-locking systems work properly, the spectrum linewidth of the ultrastable 1064 nm laser is completely transferred to the comb because the relative linewidth of the comb is at the millihertz level [8, 10]. A 578-nm light source to drive the clock transition in the Yb atom was obtained by the sum-frequency generation of a Nd:YAG laser at 1319 nm and an Yb:YAG laser at 1030 nm in waveguide periodically-poled lithium niobate (wg-PPLN) [16, 17]. The dominant part of the laser power at 578 nm was delivered to ^{171}Yb atoms trapped in an optical lattice via a 40-m-long noise-canceled SMF and the laser then irradiated the Yb atoms. The residual laser power was delivered to the comb via a noise-canceled 40-m SMF to detect the beat note between the 578-nm laser and the second harmonic generation of the comb modes in the 1156-nm region. To stabilize the 578-nm slave laser, the beat frequency was phase-locked to the RF reference (30 MHz) using an acousto-optic modulator (AOM). The servo bandwidth was estimated to be 100 kHz from servo bumps in the locked beat spectrum. The control signal for the AOM was also fed back to a PZT and a temperature controller in the 1319-nm Nd:YAG laser after integration to maintain the frequency of the AOM within the operating range.

Figure 2 shows the spectral density of the phase noise as regards the phase locked beat note between the comb and the 578-nm laser. The RMS phase ϕ is 0.23 rad, which is estimated by integrating the phase noise from 30 mHz to 2 MHz. The corresponding energy concentration ($\exp(-\phi^2)$) is approximately 95%. In addition, from our previous study [10], our comb can transfer the laser linewidth of a master laser to a comb mode at another wavelength with a coherent energy concentration peak of more than 90%. These high energy concentrations mean that this system efficiently transfers the linewidth of the master laser (few Hz) to the slave laser, and imply that the slave laser does not have an unexpected energy that could affect high-resolution spectroscopy around the coherent peak.

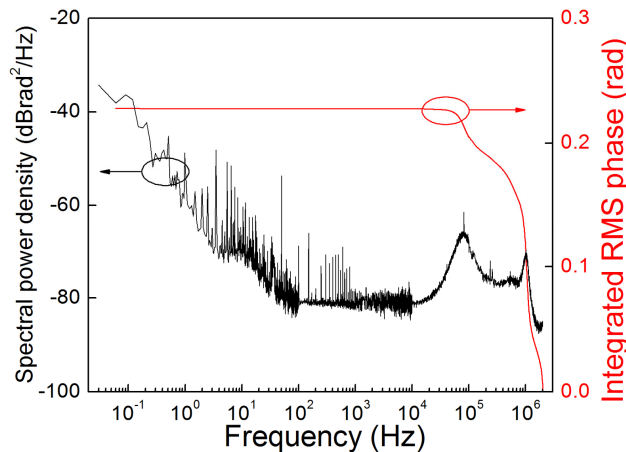


Fig. 2. Spectral density of the phase noise and the integrated RMS phase of the phase locked beat note between the 578 nm slave laser and the comb.

We employed the following experimental procedure to determine the excitation fraction of the clock transition using the achieved narrow-linewidth clock laser. We irradiated the 1S_0 ground state atoms in the optical lattice with the 578-nm clock laser pulse (width: 40 ms, intensity: $4.6 \mu\text{W}/\text{cm}^2$) to excite the atoms to the 3P_0 state. We then observed the fluorescence I_S from the atoms in the 1S_0 ground state (unexcited atoms) by driving the strong $^1S_0 - ^1P_1$

transition at 399 nm. Note that all the ground state atoms were dispersed after this process. We irradiated the 578-nm clock laser again to de-excite the atoms from the 3P_0 state to the ground state. We then observed the fluorescence I_p from the once excited atoms. Thus we could calculate the excitation fraction that canceled out the shot-to-shot atom number fluctuations as $I_p/(I_s + I_p)$. The typical excitation fraction as a function of clock laser frequency is shown in Fig. 3. We were able to observe two components in the spectrum that corresponded to the π transitions between the nuclear Zeeman sublevels of ^{171}Yb in a DC magnetic field. The clock transition spectra had almost Fourier limited linewidths of ~ 20 Hz determined by the irradiation duration (40 ms) of the clock laser to the atoms. The duration is adjusted that the slow frequency jitter of the clock laser does not exceed the Fourier limited linewidth of the atom spectra. The transfer system itself has a capability to transfer narrower linewidth of a laser to other wavelengths at millihertz level [8, 10]. We will be able to observe narrower atomic spectra if the frequency fluctuation of the master laser is further reduced. We also expect that several technical improvements of the atomic spectroscopy, such as a perfect alignment of the clock laser with the trap potentials made by the lattice laser, should help to observe further narrow linewidth on the atomic spectra. The present result shows the effectiveness of the laser linewidth transfer to clock lasers from frequency stabilized lasers operated at different wavelengths using a frequency comb with a narrow linewidth.

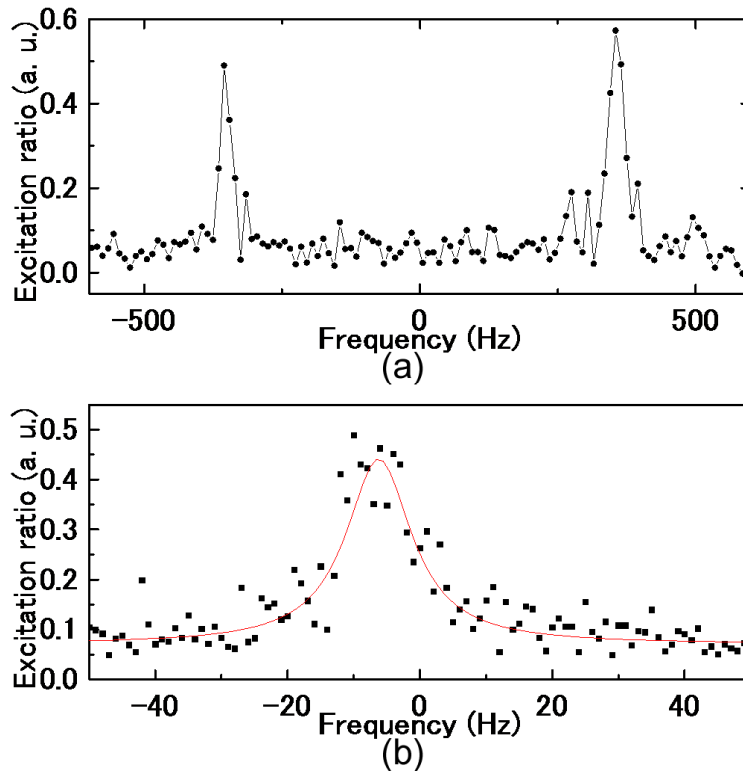


Fig. 3. Observed spectra of ^{171}Yb clock transition ($^1S_0 - ^3P_0$) with 578 nm laser based on 1064-nm cavity. (a) Two components in the spectrum that corresponded to the π transitions between the nuclear Zeeman sublevels of ^{171}Yb in a DC magnetic field. (b) An enlarged spectrum of the two components and the fitted Lorentzian.

3. Conclusion

We demonstrated ^{171}Yb clock transition spectroscopy using a novel laser system. The system does not involve an extra ultrastable cavity at the clock transition wavelength, or any laser pre-stabilization. This system can yield ultrastable lasers at arbitrary wavelengths using an

ultrastable master cavity at an arbitrary wavelength. We use a clock laser system employing linewidth transfer as a key tool for our optical lattice clocks (Yb and Sr). First, this system reduces the number of cavities in optical clock systems. Second, the system, which includes a 1064-nm ultrastable laser and a narrow linewidth frequency comb, can be continuously and easily operated for long periods of time rather than employing many cavity systems. In addition, the frequency of an optical clock can be compared with Universal Coordinated Time (UTC) or another optical clock by using information such as the f_{rep} of the comb in the system. To discuss the redefinition of the SI second, measurements of the frequency ratios between optical clocks, which do not include the current microwave-based frequency standards, have been a crucial requirement. The linewidth transfer technique is a promising way to compare optical clocks at different wavelengths and to improve the frequency stability of optical frequency ratio measurements. Namely, the frequency fluctuation of a 1064-nm master laser will be greatly suppressed by performing synchronized measurements of Sr and Yb lattice clocks using linewidth transferred clock lasers (698 and 578 nm). The relative frequency stability of our narrow linewidth comb is 3×10^{-16} at a 1-s averaging time, and it improved to 3×10^{-18} with 1000-s averaging [10], which supports comparison with highly accurate optical clocks.

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